

e-Security Authentication and Authorization Using Advanced Cryptomathic Systems

¹Manisha D. Choudhary, ²Manjusha Choudhary and ³Chounste G.B

¹H.O.D Mayur I.T & M. College, Nanded

²Yeshwant College, Nanded

³N.E.S. Science College, Nanded

e-mail: nishachoudhary98@gmail.com, ishaelanza@gmail.com, chounste@rediffmail.com

Abstract—E-Based systems and computer networks are ever-present in the modern world. With applications spanning e-commerce, WLANs, healthcare and governmental organizations, banking, among others. The secure transfer of information has therefore become a critical area of research, development, and investment.

Keywords: e-security, Authentication, authorization, Cryptomathic

I. INTRODUCTION

In last few years a rapid growth of internet browsing & surfing has taken place in all over world. The concept of internet is most important for internet users. Internet is great source of information but viruses & hacker are main problem of internet.

So the present study consists of providing e-Security on WWW data in encrypted form identified with digital Cryptomathic techniques.

E-Security solutions aim to provide five important services:

- i) Authentication of user's
- ii) Integrity
- iii) Confidentiality of communication
- iv) Availability of business services
- v) Non-repudiation of transactions.

Most e-security solutions that are provided by the literature use two main cryptographic techniques: public key cryptosystems and digital signatures.

Organizations face three types of economic impact as possible results of malicious attacks targeting their information systems: immediate, short term, and midterm economic impacts. The core areas of security of e-systems and computer networks are authentication of users, system integrity, confidentiality of communication, availability of business services, and non-repudiation of transactions.

So In this talk we argue that traditional approaches for authorization and access control in computer systems are not secure. Cryptomathic methods.

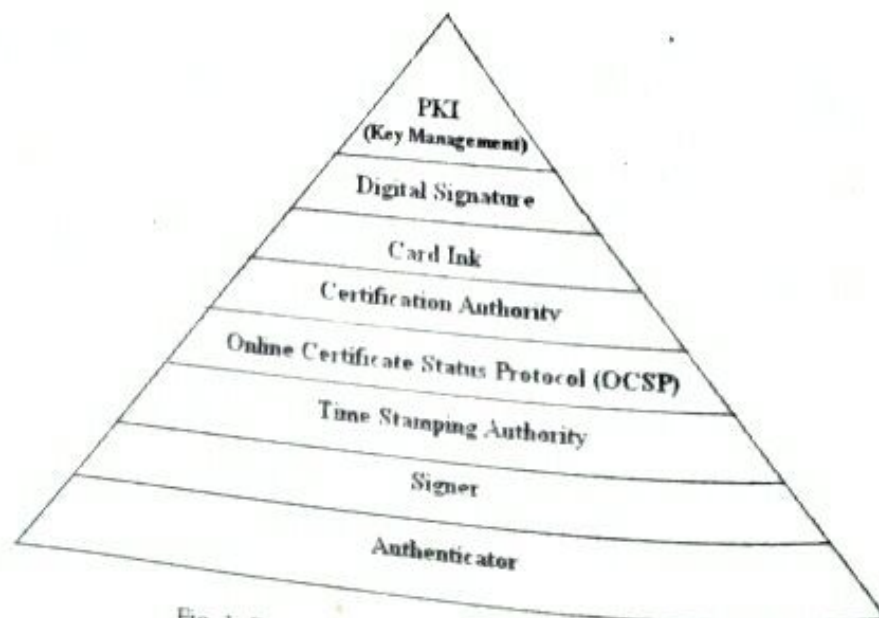


Fig. 1. e

CLOUD COMPUTING AND SECURITY THREATS IN VIRTUALIZATION

ISBN: 978-93-83414-51-2

Dr. N. V. Kalyankar
Principal

Yeshwant Mahavidyalaya
Nanded, India
nkalyankar@yahoo.com

Mr. G. B. Chounste
Assistant Professor

Department of CS & IT
Yeshwant Mahavidyalaya
Nanded., India
chounste@gmail.com

Mr.S.S Bolwar
Assistant Professor

Department of CS & IT
Yeshwant Mahavidyalaya
Nanded, India
sushant_bolwar@yahoo.co.in

ABSTRACT:

Cloud computing provides a shared pool of computing resources that can be rapidly and elastically provisioned and released based on users' demand to serve a wide and constantly expanding range of information processing needs. Due to its tremendous advantages this technology is maturing rapidly and is being adopted in many applications including government, business, and education. Virtualisation is the key concept for the success of cloud computing, this advantageous technology also contributes to security issues. Hence study of these threats is also crucial.

Keywords: Cloud Computing, Virtualization, Security

INTRODUCTION:

Cloud computing is a colloquial expression used to describe a variety of different types of computing concepts that involve a large number of computers that are connected through a real-time communication network (typically the Internet). In science, cloud computing is a synonym for distributed computing over a network and means the ability to run a program on many connected computers at the same time. The phrase is also, more commonly, used to refer to network based services which appear to be provided by real server hardware, but which in fact are served up by virtual hardware, simulated by software running on one or more real machines. Such virtual servers do not physically exist and can therefore be moved around and scaled up (or down) on the fly, without affecting the end user - arguably, rather like a cloud.[3]

Virtualization is extremely well suited to a dynamic cloud infrastructure because it provide important advantage in sharing, manageability and isolation(multiple users and applications can share physical resources without effecting one another). Virtualization allows a set of underutilized physical servers to be consolidated into smaller number of more fully utilized physical server, contributing to significant cost saving. The sever virtualization is accomplished by the use of a hypervisor. The hypervisor allow guest operating system running on the virtual machine, to function as if it is solely in control of the hardware. Unaware that other guests are sharing it. Each guest operating system is protected from the other. Hypervisors are becoming virtualization layer on client and server systems in practice cloud service can be grouped into three categories: software as a service (SaaS), Platform as a service (PaaS) and Infrastructure as a service (IaaS). IaaS is a evolution of virtual private server offering. It depends heavily on the underling infrastructure: Virtualization. Virtualization software is usually called hypervisor. It allows a single physical server to host many guest virtual machines (VM) and guest VMs are provided to client as the cloud computing service. The guest virtual machine enables individual developer and companies to deploy their own operating system and applications without the cost and complexity of buying servers.[8]

Architectural layers of cloud computing:

Cloud computing can illustrate services being provided at any of the usual layers from hardware to applications in "Fig. 1". Cloud service providers tend to offer services that can be grouped into three categories: software as a service, platform as a service, and infrastructure as a service.[2],[10] These categories group together the various layers illustrated in Figure 1.